

A Tight Analysis of Khatri-Rao Oblivious Subspace Embeddings

Lorenzo Beretta¹ and Cameron Musco²

¹IBM, lorenzo2beretta@gmail.com

²UMass Amherst, cameron.musco@gmail.com

Abstract

We study random sketching matrices with Khatri-Rao structure. In particular, we consider the Khatri-Rao product (i.e., column-wise tensor product) $A_1 \odot \cdots \odot A_d \in \mathbb{R}^{(n_1 \cdots n_d) \times m}$ of random matrices $A_i \in \mathbb{R}^{n_i \times m}$ whose columns are isotropic, independent and sub-Gaussian (e.g., Gaussian matrices). Khatri-Rao sketching matrices are widely applied in randomized algorithms for linear algebraic computation and data analysis, when the input data has tensor structure that allows for fast multiplication with $A_1 \odot \cdots \odot A_d$. However, existing theory is not able to fully explain their performance in practice. In particular, despite significant attention, our best bounds for the important *oblivious subspace embedding* property with Khatri-Rao matrices lag behind what is achievable with standard unstructured matrices.

For embedding a k -dimensional subspace to $(1 \pm \varepsilon)$ error, Bujanović et al. [6] prove that sketching dimension $m = O(k^{3/2}/\varepsilon^2)$ suffices in the special case of $d = 2$. Their dependence on k is weaker than the tight bound of $O(k/\varepsilon^2)$ known for unstructured sub-Gaussian sketching matrices. In this work, we close this gap, showing that $m = \tilde{O}(k/\varepsilon^2)$ suffices for subspace embedding with a Khatri-Rao sketching matrix with any fixed order d . Our proof is simple, leveraging just two basic properties of the Khatri-Rao sketching distribution: 1) the columns of $A_1 \odot \cdots \odot A_d \in \mathbb{R}^{(n_1 \cdots n_d) \times m}$ are independent and isotropic, and 2) each column of $A_1 \odot \cdots \odot A_d \in \mathbb{R}^{(n_1 \cdots n_d) \times m}$ satisfies a weak Johnson-Lindenstrauss type moment property.

1 Introduction

Random sketching is a key tool in work on fast algorithms for linear algebraic computation, large-scale data analysis, and low-memory streaming computation [16, 15, 24]. Given an input matrix $X \in \mathbb{R}^{n \times p}$, we generate a *random sketching matrix* $\Psi \in \mathbb{R}^{n \times m}$, and compute the sketch $\Psi^T X \in \mathbb{R}^{m \times p}$. $\Psi^T X$ serves as a lossy compression of X , that can be used to approximately solve a breadth of problems, such as low-rank approximation [11], linear regression [19], clustering [10], eigenvalue problems [21], and beyond.

In many applications, X is known to be structured – e.g., it may be sparse or may be given as the tensor product of several small matrices. In such applications, it is desirable to use a sketching matrix Ψ that also has structure, that facilitates fast computation of $\Psi^T X$. In applications where X has tensor structure, a key tool in the literature is the *Khatri-Rao random sketch* [17, 2, 1, 20, 9, 3, 6, 18, 8, 12].

Definition 1.1 (Khatri-Rao product). Let $A \in \mathbb{R}^{n_a \times m}$, $B \in \mathbb{R}^{n_b \times m}$ have columns $A = [a_1, \dots, a_m]$ and $B = [b_1, \dots, b_m]$. Their *Khatri-Rao product* is the column-wise Kronecker product

$$A \odot B := [a_1 \otimes b_1, \quad \dots, \quad a_m \otimes b_m] \in \mathbb{R}^{(n_a \cdot n_b) \times m}.$$

In this work we consider a sketching matrix $\Psi = \frac{1}{\sqrt{m}} A_1 \odot \cdots \odot A_d$ that is the Khatri-Rao product of sketching matrices $A_i \in \mathbb{R}^{n_i \times m}$, appropriately scaled. We will focus on the common case when all A_i have independent sub-Gaussian columns, which of course includes independent standard Gaussian matrices. Although another important variant is the *Tensor Sketch* of Pham and Pagh [17], where the A_i are sparse Count-Sketch matrices.

When the columns of $X \in \mathbb{R}^{(n_1 \cdots n_d) \times d}$ are also d -order tensor products, one can compute $\Psi^T X$ in $O((n_1 + \cdots + n_d) \cdot dm)$ time, as compared to $O(n_1 \cdots n_d \cdot dm)$ time for a dense unstructured random projection. This represents an exponential savings in cost. For this reason, Khatri-Rao sketches have found applications in numerous settings, ranging from contour integral-based eigensolvers [6], to compression algorithms for massive tensors and tensor networks, arising e.g., in quantum physics [7, 12], to approximation algorithms for polynomial kernels [2, 1], and more [14, 8, 9, 5]

1.1 Oblivious Subspace Embeddings from Khatri-Rao Sketches

Given their importance, significant work has studied theoretical error bounds for Khatri-Rao random sketches. Much of this work has focused on two related guarantees: (ε, δ) -*Johnson-Lindenstrauss embedding* and (ε, δ, k) -*oblivious subspace embedding (OSE)*. For the first, it has been shown by Ahle et al. [1] that for any $\varepsilon, \delta \in (0, 1)$ and any fixed $x \in \mathbb{R}^{n_1 \cdots n_d}$, $(1 - \varepsilon) \|x\|_2 \leq \|\Psi^T x\|_2 \leq (1 + \varepsilon) \|x\|_2$ with probability at least $1 - \delta$ when the sketching dimension is at least

$$m = O_d \left(\frac{\log(1/\delta)}{\varepsilon^2} + \frac{\log(1/\delta)^d}{\varepsilon} \right), \quad (1)$$

where $O_d(\cdot)$ hides $2^{O(d)}$ factors. Similar bounds appear e.g., in [20, 13, 3, 12]. As compared to dense unstructured random projections, which achieve the same bound with $m = O(\log(1/\delta)/\varepsilon^2)$, the main loss in using a Khatri-Rao structured sketch is the dependence on $\log(1/\delta)^d$. This work focuses on oblivious subspace embeddings, defined below.

Definition 1.2 (Oblivious subspace embedding). Let $\varepsilon, \delta \in (0, 1)$ and let $k \leq N$ be positive integers. A random matrix $\Psi \in \mathbb{R}^{N \times m}$ is an (ε, δ, k) -*oblivious subspace embedding (OSE)* if, for every k -dimensional subspace $V \subseteq \mathbb{R}^N$,

$$\mathbb{P}_{\Psi} \left[(1 - \varepsilon) \|x\|_2 \leq \|\Psi^T x\|_2 \leq (1 + \varepsilon) \|x\|_2 \text{ for all } x \in V \right] \geq 1 - \delta.$$

The OSE property is a key tool in deriving theoretical analysis for sketched linear regression, low-rank approximation, and many other linear algebraic problems [19, 24]. It can be derived from a Johnson-Lindenstrauss bound via a standard net argument – it suffices for Ψ to simultaneously preserve the norm of $2^{O(k)}$ points that form a covering of the subspace. To ensure this with probability at least $1 - \delta$, one sets the failure probability for embedding to $\delta/2^{O(k)}$ and applies a union bound. Unfortunately, following this approach and applying the bound of (1) gives embedding dimension for Khatri-Rao (ε, δ, k) -OSEs of:

$$m = O_d \left(\frac{\log(2^{O(k)}/\delta)}{\varepsilon^2} + \frac{\log(2^{O(k)}/\delta)^d}{\varepsilon} \right) = O_d \left(\frac{k \log(1/\delta)}{\varepsilon^2} + \frac{(k + \log(1/\delta))^d}{\varepsilon} \right). \quad (2)$$

See e.g., [18, 12] for examples of works that derive and apply such a bound. As compared to standard unstructured embeddings, which achieve $O(k \log(1/\delta)/\varepsilon^2)$, the bound has a significantly suboptimal dependence on k , even e.g., when $d = 2$, where the bound is quadratic, rather than linear in k .

Recently, Bujanović, Grubišić, Kressner, and Lam [6] partially resolved this gap, focusing on the special case of $d = 2$. In this setting, they give an improved bound of

$$m = O \left(\frac{k^{3/2} + k \log(1/\delta)}{\varepsilon^2} + \frac{k^{1/2} \log(1/\delta)^2}{\varepsilon} \right). \quad (3)$$

However, this bound still loses a $\sqrt{k}$ factor as compared to standard unstructured sketches. Camaño, Eperly, Meyer, and Tropp [8] show that the gap can be closed if one instead asks for a weaker *oblivious subspace injection* property, which is sufficient e.g., in the analysis of some low-rank approximation algorithms. However, deriving tight bounds for oblivious subspace embedding with Khatri-Rao sketches, or showing that existing gaps from unstructured sketches are inherent, has remained open.

1.2 Our Results

We resolve this question, showing that Khatri-Rao random sketches can indeed nearly match unstructured sketches by achieving OSEs with a nearly linear dependence on the subspace dimension k .

Theorem 1.3 (Main Theorem). *Let $\varepsilon, \delta \in (0, 1)$, $n_1, \dots, n_d \geq 2$ and $k \leq N = \prod_{i=1}^d n_i$. Let $A_i \in \mathbb{R}^{n_i \times m}$ be random matrices whose columns are independent, isotropic and have constant sub-Gaussian norm. Let $\Psi = \frac{1}{\sqrt{m}} A_1 \odot \dots \odot A_d \in \mathbb{R}^{N \times m}$. Then Ψ is an (ε, δ, k) -OSE provided that*

$$m \geq d^{O(d)} \cdot \frac{k}{\varepsilon^2} \cdot \left(\log \left(\frac{k}{\varepsilon \delta} \right) \right)^{d+1}.$$

Fixing ε, δ to be constants, in the case of $d = 2$, our bound improves on the existing state-of-the-art bound in (3) by replacing a $\sqrt{k}$ with a $\log(k)^3$. For $d > 2$, the best known prior bound is (2), which, we improve by replacing a k^d with a $k \log(k)^{d+1}$.

Proof Overview. Let $\Phi = A_1 \odot \dots \odot A_d$ be the unnormalized sketch, so $\Psi = \frac{1}{\sqrt{m}} \Phi$. The proof of our main result is elementary and uses just two simple facts about Khatri-Rao random sketches:

1. **The columns of $\Phi \in \mathbb{R}^{N \times m}$ are independent and isotropic.** When the columns of $A_1, \dots, A_d$ are independent and isotropic, this is straightforward. Indeed, the columns of Φ are independent since they are tensor products of different independent columns of $A_1, \dots, A_d$. They are isotropic since they are tensorizations of the independent isotropic columns of $A_1, \dots, A_d$.
2. **The columns of $\Phi \in \mathbb{R}^{N \times m}$ satisfy a weak Johnson-Lindenstrauss type moment property.** In particular, for any $U \in \mathbb{R}^{N \times k}$ with orthonormal columns, we show that, with probability at least $1 - \delta$,

$$\|U^T \Phi_{*,i}\|_2^2 \leq 2^{O(d)} \cdot \|U\|_F^2 \log(1/\delta)^d = 2^{O(d)} \cdot k \log(1/\delta)^d.$$

This bound is similar in spirit and proven using similar tools to existing Johnson-Lindenstrauss bounds for Khatri-Rao sketches, like (1). In fact, a very similar bound is shown in [4], although they give bounds that are weaker in our parameter regime.

With these two facts in hand, we prove our result using a standard matrix Chernoff bound [22]. In particular, Ψ is an (ε, δ, k) -OSE if for any orthonormal $U \in \mathbb{R}^{N \times k}$, with probability at least $1 - \delta$, for all unit-norm $x \in \mathbb{R}^k$,

$$(1 - \varepsilon) \leq x^T U^T \Psi \Psi^T U x \leq (1 + \varepsilon). \quad \text{Equivalently if: } \|U^T \Psi \Psi^T U - I\|_{\text{op}} \leq \varepsilon.$$

By the observation that the columns of Φ are isotropic, we have $\mathbb{E}[U^T \Psi \Psi^T U] = I$. So our bound amounts to showing that $\|U^T \Psi \Psi^T U - \mathbb{E}[U^T \Psi \Psi^T U]\|_{\text{op}} \leq \varepsilon$. Writing $U^T \Psi \Psi^T U$ as an average of m independent outer products

$$U^T \Psi \Psi^T U = \frac{1}{m} \sum_{i=1}^m U^T \Phi_{*,i} \Phi_{*,i}^T U,$$

this deviation is exactly what a matrix Chernoff bound allows us to bound. We just need an upper bound R^2 on the squared norms $\|U^T \Phi_{*,1}\|_2^2, \dots, \|U^T \Phi_{*,m}\|_2^2$. But this is given (with high enough probability) by our weak Johnson-Lindenstrauss type bound. In particular, union bounding over all m columns, we know that, with high probability, all squared norms are simultaneously bounded by $R^2 = O(k \log(m/\delta)^d)$. We can simply *truncate* the vectors in the low-probability event that the bound fails, giving a small error which is absorbed into the final OSE bound.

Ultimately, just applying the standard matrix Chernoff bound, this gives that an embedding dimension of m suffices as long as

$$m \gtrsim \frac{R^2 \log(k/\delta)}{\varepsilon^2} = \frac{k \log(m/\delta)^d \log(k/\delta)}{\varepsilon^2},$$

which holds for $m = O_d(\varepsilon^{-2} \cdot k \log(k/(\varepsilon \delta))^{d+1})$. This gives the bound stated in Theorem 1.3.

2 Proof of the Main Theorem

Throughout, $\Phi = A_1 \odot \cdots \odot A_d \in \mathbb{R}^{N \times m}$ and $\Psi = \frac{1}{\sqrt{m}}\Phi$ is the Khatri–Rao sketch of Theorem 1.3, with $N = \prod_{i=1}^d n_i$. We fix an arbitrary k -dimensional subspace and denote by $U \in \mathbb{R}^{N \times k}$ a matrix whose columns form an orthonormal basis of that subspace.

In Section 2.1 we prove that $\Phi_{*,1} \dots \Phi_{*,m}$ are *isotropic* and statistically independent. In Section 2.2 we prove a *tail bound* bounding the probability that $\|U^\top \Phi_{*,i}\|_2^2$ is much larger than its expectation. In Section 2.3, we leverage these two properties along with the matrix Chernoff bound to prove Theorem 1.3.

2.1 The columns of Φ are isotropic and independent

The following lemma states that a Kronecker product of independent isotropic vectors is again isotropic.

Lemma 2.1. *Let $g_i \in \mathbb{R}^{n_i}$ for all $i \in [d]$ be statistically independent, isotropic, random vectors. Then, $\phi = g_1 \otimes \cdots \otimes g_d$ is isotropic, that is, $\mathbb{E}[\phi\phi^\top] = I$.*

Proof. It suffices to treat two factors, since the general case then follows by induction on d . Let $g \in \mathbb{R}^{n_1}$ and $h \in \mathbb{R}^{n_2}$ be independent and isotropic. Their Kronecker product $g \otimes h$ has entries $(g \otimes h)_{(i,j)} = g_i h_j$, so

$$\mathbb{E}[(g \otimes h)_{(i,j)} (g \otimes h)_{(i',j')}] = \mathbb{E}[g_i g_{i'}] \mathbb{E}[h_j h_{j'}] = \mathbf{1}\{i = i'\} \mathbf{1}\{j = j'\},$$

using independence for the factorization and isotropy of g and h for each factor. Thus, $\mathbb{E}[(g \otimes h)(g \otimes h)^\top] = I_{n_1 n_2}$. Applying this identity repeatedly to $g_1 \otimes \cdots \otimes g_d$ shows that ϕ is isotropic. $\square$

Observation 2.2. *The columns of Φ are statistically independent as they are defined as Kronecker products of statistically independent vectors.*

2.2 Upper tail bound for $\|U^\top \Phi_{*,i}\|_2^2$

Isotropy of $\Phi_{*,i}$ implies that $\mathbb{E}[\|U^\top \Phi_{*,i}\|_2^2] = k$. The following lemma states that $\|U^\top \Phi_{*,i}\|_2^2$ exceeds k only by a factor of $O(\log(1/\delta)^d)$ at failure probability δ .

Proposition 2.3. *Let $g_i \in \mathbb{R}^{n_i}$ for all $i \in [d]$ be independent, isotropic random vectors, and let $\phi = g_1 \otimes \cdots \otimes g_d \in \mathbb{R}^N$. Then there is a constant $C = 2^{O(d)}$ such that for every $t \geq 1$ and every $U \in \mathbb{R}^{N \times k}$ with orthonormal columns,*

$$\mathbb{P}\left[\|U^\top \phi\|_2^2 > C k t\right] \leq \exp\left(-t^{1/d}\right).$$

Writing $U = [U_1, \dots, U_k]$ and $y := U^\top \phi \in \mathbb{R}^k$, each coordinate $y_j = \langle U_j, \phi \rangle$ is a degree- d polynomial in the entries of the independent factors $g_1, \dots, g_d$. Concretely, reshaping the vector U_j into an order- d tensor $T = T^{(j)}$ with $T_{i_1 \dots i_d} = (U_j)_{(i_1, \dots, i_d)}$, we may write

$$y_j = \sum_{i_1, \dots, i_d} T_{i_1 \dots i_d} (g_1)_{i_1} \cdots (g_d)_{i_d}.$$

We would like to bound the moments of y_j . For $d = 1$ this is standard: a linear combination of independent sub-Gaussian coordinates has moments growing like $\sqrt{p}$, which we quote in the form we will use.

Imported Theorem 1 ([23, Lemma 3.4.2 and Prop. 2.6.1]). Let $h = (h_1, \dots, h_n)$ be a random vector whose entries are independent and have constant sub-Gaussian norm. Then, h has constant sub-Gaussian norm, and there exists a constant $c \geq 1$ such that for all $a \in \mathbb{R}^n$ and $p \geq 2$,

$$\|\langle a, h \rangle\|_{L_p} \leq c \cdot \sqrt{p} \|a\|_2.$$

The following lemma extends the p -th moment bound on y_j given by Imported Theorem 1 to $d > 1$.

Lemma 2.4 (Moment bound for sub-Gaussian chaoses). *Let $c \geq 1$ be the constant of Imported Theorem 1. For every order- d tensor T , every collection of independent, isotropic, sub-Gaussian vectors $g_1, \dots, g_d$ with constant sub-Gaussian norm, and every $p \geq 2$,*

$$\left\| \sum_{i_1, \dots, i_d} T_{i_1 \dots i_d} (g_1)_{i_1} \cdots (g_d)_{i_d} \right\|_{L_p} \leq c^d p^{d/2} \|T\|_F.$$

Proof. We induct on d . When $d = 1$ the form is the linear combination $\langle T, g_1 \rangle$, and Imported Theorem 1 gives $\|\langle T, g_1 \rangle\|_{L_p} \leq c\sqrt{p} \|T\|_2$, which is the claim.

For the inductive step, group the sum according to its last index. Writing S for the order- d form and $W_{i_d} := \sum_{i_1, \dots, i_{d-1}} T_{i_1 \dots i_d} (g_1)_{i_1} \cdots (g_{d-1})_{i_{d-1}}$ for the order- $(d-1)$ form obtained by fixing the last index, we have $S = \sum_{i_d} (g_d)_{i_d} W_{i_d} = \langle W, g_d \rangle$. Crucially, the vector $W = (W_{i_d})_{i_d}$ depends only on $g_1, \dots, g_{d-1}$, hence is independent of g_d . Conditioning on $g_1, \dots, g_{d-1}$ and applying Imported Theorem 1 in the block g_d ,

$$\|S\|_{L_p} \leq c \sqrt{p} \|W\|_2 \|g_d\|_{L_p}.$$

It remains to bound $\|W\|_2 \|g_d\|_{L_p}$. Squaring and using the triangle inequality in $L_{p/2}$ (available since $p \geq 2$),

$$\|W\|_2^2 \|g_d\|_{L_p}^2 = \left\| \sum_{i_d} W_{i_d}^2 \right\|_{L_{p/2}} \leq \sum_{i_d} \|W_{i_d}\|_{L_p}^2.$$

Each W_{i_d} is a multilinear form of order $d-1$ with coefficient tensor T_{*, i_d} , so the inductive hypothesis bounds $\|W_{i_d}\|_{L_p} \leq c^{d-1} p^{(d-1)/2} \|T_{*, i_d}\|_F$. Summing over i_d and using $\sum_{i_d} \|T_{*, i_d}\|_F^2 = \|T\|_F^2$ gives $\|W\|_2 \|g_d\|_{L_p} \leq c^{d-1} p^{(d-1)/2} \|T\|_F$. Combining the two displays yields $\|S\|_{L_p} \leq c^d p^{d/2} \|T\|_F$, completing the induction. $\square$

We can now prove the tail bound.

Proof of Proposition 2.3. Applying Lemma 2.4 to y_j , whose coefficient tensor $T^{(j)}$ is just the reshaping of U_j and so has Frobenius norm $\|U_j\|_2 = 1$, gives $\|y_j\|_{L_{2q}} \leq c^d (2q)^{d/2}$ for every $q \geq 1$. Combining these bounds and applying triangle inequality we obtain

$$\|y\|_2^2 \|y\|_{L_{2q}}^2 = \left\| \sum_{j=1}^k y_j^2 \right\|_{L_q} \leq \sum_{j=1}^k \|y_j\|_{L_{2q}}^2 \leq k c^{2d} (2q)^d.$$

Markov's inequality applied to the $2q$ -th moment of $\|y\|_2$ therefore gives, for every threshold $s > 0$,

$$\mathbb{P} \left[\|y\|_2^2 \geq s k \right] \leq \frac{\|y\|_2^{2q} \|y\|_{L_{2q}}^{2q}}{(sk)^q} \leq \left(\frac{c^{2d} (2q)^d}{s} \right)^q.$$

It remains to choose the moment order q . Set $C := e^2 \cdot c^{2d} = 2^{O(d)}$, and fix a threshold of the form $s = C t$ with $t \geq 1$. Taking $q = \frac{1}{2} t^{1/d}$, we have $(2q)^d \leq t$ and

$$\frac{c^{2d} (2q)^d}{s} \leq \frac{c^{2d} t}{C t} \leq \frac{1}{e^2}.$$

Hence,

$$\mathbb{P} \left[\|y\|_2^2 \geq C k t \right] \leq e^{-2q} = \exp \left(-t^{1/d} \right).$$

□

2.3 Proof of Theorem 1.3

Our goal is to prove that with probability $1 - \delta$ we have $\|\Psi^\top x\|_2 = (1 \pm \varepsilon)\|x\|_2$ for all $x \in \text{Span}(U)$. Up to constant factors in ε , we can rephrase our desiderata as

$$\mathbb{P} \left[(1 - \varepsilon)I_k \preceq U^\top \Psi \Psi^\top U \preceq (1 + \varepsilon)I_k \right] \geq 1 - \delta.$$

Let $\Psi = \frac{1}{\sqrt{m}} [\phi_1 \mid \cdots \mid \phi_m]$, so that $\phi_1, \dots, \phi_m$ are the columns of Φ , and set $z_i := U^\top \phi_i$. Then

$$U^\top \Psi \Psi^\top U = \frac{1}{m} \sum_{i=1}^m z_i z_i^\top,$$

is the average of m statistically independent rank-one matrices. Our goal is to show

$$\mathbb{P} \left[(1 - \varepsilon)I_k \preceq \frac{1}{m} \sum_{i=1}^m z_i z_i^\top \preceq (1 + \varepsilon)I_k \right] \geq 1 - \delta. \quad (4)$$

To prove Equation (4) we will use the following matrix concentration bounds.

Imported Theorem 2 (Matrix Chernoff bounds [22]). Let $Y_1, \dots, Y_m \in \mathbb{R}^{k \times k}$ be independent, positive semidefinite, and satisfy $Y_i \preceq R^2 I_k$ almost surely. Write $\mu_{\min}$ and $\mu_{\max}$ for the smallest and largest eigenvalues of $\sum_i \mathbb{E}[Y_i]$. Then for every $t \in (0, 1)$,

$$\begin{aligned} \mathbb{P} \left[\lambda_{\max} \left(\sum_i Y_i \right) \geq (1 + t) \mu_{\max} \right] &\leq k \exp \left(-\frac{t^2 \mu_{\max}}{3R^2} \right), \\ \mathbb{P} \left[\lambda_{\min} \left(\sum_i Y_i \right) \leq (1 - t) \mu_{\min} \right] &\leq k \exp \left(-\frac{t^2 \mu_{\min}}{2R^2} \right). \end{aligned}$$

A truncation argument. Unfortunately, the quality of the bound in Imported Theorem 2 depends on a uniform bound on the spectral radius of all $z_i z_i^\top$ for $i \in [m]$. However, the terms $z_i z_i^\top$ might have arbitrarily large spectral radius. We obviate this issue via a truncation argument. Let

$$R^2 := C k (\log(m/\delta))^d,$$

which is the tail threshold of Proposition 2.3 at failure probability δ/m , and accordingly $C = 2^{O(d)}$. Then, set $\tilde{z}_i := z_i \mathbf{1}_{\{\|z_i\|_2 \leq R\}}$ and $Y_i := \tilde{z}_i \tilde{z}_i^\top$. By construction $0 \preceq Y_i \preceq R^2 I_k$. By the choice of R and Proposition 2.3, each $i \in [m]$ satisfies $\|z_i\|_2 > R$ with probability at most δ/m , so a union bound over $i \in [m]$ shows that with probability at least $1 - \delta$ no z_i is truncated. Thus the following observation holds.

Observation 2.5. *With probability at least $1 - \delta$ we have $\frac{1}{m} \sum_i Y_i = U^\top \Psi \Psi^\top U$.*

Our next goal is to prove that

$$\mathbb{P} \left[(1 - \varepsilon)I_k \preceq \frac{1}{m} \sum_{i=1}^m Y_i \preceq (1 + \varepsilon)I_k \right] \geq 1 - \delta.$$

We will do this in two steps: first, we prove that the expectation of $\frac{1}{m} \sum_{i=1}^m Y_i$ is nearly the identity, and then we prove that $\sum_{i=1}^m Y_i$ concentrates about its expectation using Imported Theorem 2.

Lemma 2.6. $(1 - O(\varepsilon))I_k \preceq \mathbb{E} \left[\frac{1}{m} \sum_{i=1}^m Y_i \right] \preceq I_k.$

Proof. By Lemma 2.1, we have $\mathbb{E} [z_i z_i^\top] = I_k$, so

$$\mathbb{E} [Y_i] = I_k - \mathbb{E} [z_i z_i^\top \mathbf{1}_{\{\|z_i\|_2 > R\}}],$$

and the correction term $\mathbb{E} [z_i z_i^\top \mathbf{1}_{\{\|z_i\|_2 > R\}}]$ is positive semidefinite, hence $\mathbb{E} [Y_i] \preceq I_k$. Writing $W := \|z_i\|_2^2$, the operator norm of the correction term is at most $\mathbb{E} [W \mathbf{1}_{\{W > R^2\}}]$. Read as a tail bound, Proposition 2.3 states $\mathbb{P} [W > t] \leq \exp(-(t/(Ck))^{1/d})$ for $t \geq Ck$. Setting $s_0 := (R^2/Ck)^{1/d} = \log(m/\delta)$ and substituting $s = (t/(Ck))^{1/d}$ in the layer-cake formula for $\mathbb{E} [W \mathbf{1}_{\{W > R^2\}}]$,

$$\begin{aligned} \mathbb{E} [W \mathbf{1}_{\{W > R^2\}}] &= R^2 \mathbb{P} [W > R^2] + \int_{R^2}^{\infty} \mathbb{P} [W > t] dt \\ &\leq R^2 e^{-s_0} + Ck d \int_{s_0}^{\infty} s^{d-1} e^{-s} ds \\ &= O(R^2 e^{-s_0}) \\ &= O\left(R^2 \cdot \frac{\delta}{m}\right), \end{aligned}$$

where the third line holds because $s_0 \geq 2d$, which is implied by $m \geq 2^{2d}$. Moreover, we can verify that $m \geq 2^{O(d)} \cdot \frac{k}{\varepsilon^2} \cdot (\log(\frac{k}{\varepsilon\delta}))^{d+1}$ satisfies $m = \Omega(R^2 \varepsilon^{-1})$, so $\mathbb{E} [W \mathbf{1}_{\{W > R^2\}}] = O(\varepsilon)$. Therefore, $(1 - O(\varepsilon))I_k \preceq \mathbb{E} [Y_i] \preceq I_k$. Averaging over $i \in [m]$ yields the desiderata. $\square$

Lemma 2.7. $\mathbb{P} [(1 - O(\varepsilon))I_k \preceq \frac{1}{m} \sum_i Y_i \preceq (1 + O(\varepsilon))I_k] \geq 1 - \delta.$

Proof. The matrices $Y_1, \dots, Y_m$ are independent, positive semidefinite, and $\preceq$ -bounded from above by $R^2 I_k$, so Imported Theorem 2 applies to their sum $\sum_i Y_i$. By Lemma 2.6 we have $\mu_{\max} = \lambda_{\max}(\sum_i \mathbb{E} [Y_i]) \leq m$ and $\mu_{\min} = \lambda_{\min}(\sum_i \mathbb{E} [Y_i]) \geq m(1 - O(\varepsilon)) \geq 3m/4$. Applying the two tails of Imported Theorem 2 and taking a union bound, the truncated average fails to satisfy $(1 - O(\varepsilon))I_k \preceq \frac{1}{m} \sum_i Y_i \preceq (1 + O(\varepsilon))I_k$ with probability at most $2k \cdot \exp(-\Omega(\varepsilon^2 m/R^2))$. The latter is at most δ as long as

$$m = \Omega\left(\varepsilon^{-2} R^2 \log \frac{k}{\delta}\right)$$

which is satisfied by $m \geq 2^{O(d)} \cdot \frac{k}{\varepsilon^2} \cdot (\log(\frac{k}{\varepsilon\delta}))^{d+1}$. $\square$

Finally, a union bound over the failure probabilities of Observation 2.5 and Lemma 2.7 gives a combined failure probability of at most 2δ . When both these events happen we have the desiderata, up to constant factors in ε and δ , which can be absorbed into m .

Acknowledgments. The first version of [18], available on arXiv, proved a result essentially equivalent to our main theorem using the same proof strategy. However, this result was removed from the second version after an error was identified in its proof during the review process.

We were unaware of this earlier proof attempt while developing the present work. In hindsight, we acknowledge that the main contribution of this manuscript can be viewed as providing a correct proof of the result claimed in the first version of [18].

AI Acknowledgement. We acknowledge the use of AI throughout the research process. Our proofs were developed through interactive conversation with LLMs. AI was also a valuable tool for typesetting and proofreading. The authors certify correctness of manuscript and bear responsibility for it.

References

- [1] Thomas D Ahle et al. “Oblivious sketching of high-degree polynomial kernels”. In: *Proceedings of the Fourteenth Annual ACM-SIAM Symposium on Discrete Algorithms*. SIAM. 2020, pp. 141–160.
- [2] Haim Avron, Huy L Nguyễn, and David P Woodruff. “Subspace embeddings for the polynomial kernel”. In: *Advances in Neural Information Processing Systems 27 (NuerIPS) 27* (2014).
- [3] Stefan Bamberger, Felix Krahmer, and Rachel Ward. “Johnson–Lindenstrauss embeddings with Kronecker structure”. In: *SIAM Journal on Matrix Analysis and Applications* 43.4 (2022), pp. 1806–1850.
- [4] Stefan Bamberger, Felix Krahmer, and Rachel Ward. “The Hanson–Wright inequality for random tensors”. In: *Sampling Theory, Signal Processing, and Data Analysis* 20.2 (2022), p. 14.
- [5] Zvonimir Bujanovic and Daniel Kressner. “Norm and trace estimation with random rank-one vectors”. In: *SIAM Journal on Matrix Analysis and Applications* 42.1 (2021), pp. 202–223.
- [6] Zvonimir Bujanović et al. “Subspace embedding with random Khatri-Rao products and its application to eigensolvers”. In: *IMA journal of numerical analysis* (2025), draf043.
- [7] Chris Camaño, Ethan N Epperly, and Joel A Tropp. “Successive randomized compression: A randomized algorithm for the compressed MPO-MPS product”. In: *Quantum* 10 (2026), p. 2022.
- [8] Chris Camaño et al. “Faster linear algebra algorithms with structured random matrices”. In: *arXiv preprint arXiv:2508.21189* (2025).
- [9] Ke Chen and Ruhui Jin. “Tensor-structured sketching for constrained least squares”. In: *SIAM Journal on Matrix Analysis and Applications* 42.4 (2021), pp. 1703–1731.
- [10] Michael B Cohen et al. “Dimensionality reduction for k-means clustering and low rank approximation”. In: *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC)*. 2015, pp. 163–172.
- [11] Nathan Halko, Per-Gunnar Martinsson, and Joel A Tropp. “Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions”. In: *SIAM Review* 53.2 (2011), pp. 217–288.
- [12] Cullen Haselby et al. “Fast and low-memory compressive sensing algorithms for low Tucker-rank tensor approximation from streamed measurements”. In: *Numerical Algorithms* 101.4 (2026), pp. 2567–2629.
- [13] Ruhui Jin, Tamara G Kolda, and Rachel Ward. “Faster Johnson-Lindenstrauss transforms via Kronecker products”. In: *Information and Inference: A Journal of the IMA* 10.4 (2021), pp. 1533–1562.
- [14] Daniel Kressner and Lana Perisa. “Recompression of Hadamard products of tensors in Tucker format”. In: *SIAM Journal on Scientific Computing* 39.5 (2017), A1879–A1902.
- [15] Michael W Mahoney. “Randomized algorithms for matrices and data”. In: *Foundations and Trends® in Machine Learning* 3.2 (2011), pp. 123–224.
- [16] Shanmugavelayutham Muthukrishnan. “Data streams: Algorithms and applications”. In: *Foundations and Trends® in Theoretical Computer Science* 1.2 (2005), pp. 117–236.
- [17] Ninh Pham and Rasmus Pagh. “Fast and scalable polynomial kernels via explicit feature maps”. In: *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 2013, pp. 239–247.
- [18] Arvind K Saibaba, Bhisham Dev Verma, and Grey Ballard. “Improved Analysis of Khatri-Rao Random Projections and Applications”. In: *arXiv preprint arXiv:2507.23207* (2025).
- [19] Tamas Sarlos. “Improved approximation algorithms for large matrices via random projections”. In: *Proceedings of the 47th Annual IEEE Symposium on Foundations of Computer Science (FOCS)*. 2006.
- [20] Yiming Sun et al. “Tensor random projection for low memory dimension reduction”. In: *arXiv:2105.00105* (2021).
- [21] William Swartworth and David P Woodruff. “Optimal eigenvalue approximation via sketching”. In: *Proceedings of the 55th Annual ACM Symposium on Theory of Computing (STOC)*. 2023.

- [22] Joel A Tropp. “User-friendly tail bounds for sums of random matrices”. In: *Foundations of computational mathematics* 12.4 (2012), pp. 389–434.
- [23] Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Vol. 47. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge: Cambridge University Press, 2018.
- [24] David P Woodruff. “Sketching as a tool for numerical linear algebra”. In: *Foundations and Trends® in Theoretical Computer Science* 10.1-2 (2014), pp. 1–157.